

Technical Supplement

Reassessing Recent Inequality Trends in Chile: Income, Wealth, and Survey Disagreement, 2007–2024

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This supplement records estimator definitions, validation evidence, design diagnostics, source lineage, disclosure rules, and extended results. It should be read with the paper. Restricted microdata, identifiers, exact restricted- input hashes, and private support counts are not reproduced here.

1 Guide to the empirical record

Readers looking first for the evidence behind the paper’s conclusions can use the following sequence. Table S1 reports all six EFH income and wealth levels. Table S2 reports the adjacent-wave tests, and Table S3 tests the 2017–2024 income contrast under three common EFH–CASEN conventions. Tables S4, S5, S6, and S7 address upper-tail influence, wealth-definition scope, wealth-component accounting, and survivor composition. Table S11 shows whether the EFH–CASEN income disagreement also appears in bottom-half and top-decile shares. The remaining sections document exactly how those quantities were estimated and validated.

<i>Income</i>			
Wave	Gini	B50 share	T10 share
2007	0.553 [0.536, 0.570]	0.157 [0.149, 0.165]	0.451 [0.431, 0.471]
2011	0.532 [0.521, 0.543]	0.161 [0.155, 0.167]	0.416 [0.404, 0.427]
2014	0.534 [0.517, 0.551]	0.159 [0.151, 0.168]	0.416 [0.396, 0.437]
2017	0.519 [0.507, 0.531]	0.164 [0.157, 0.170]	0.397 [0.383, 0.411]
2021	0.439 [0.427, 0.451]	0.212 [0.205, 0.220]	0.332 [0.320, 0.344]
2024	0.523 [0.494, 0.552]	0.163 [0.152, 0.175]	0.399 [0.364, 0.435]

<i>Signed non-pension wealth</i>			
Wave	Gini	B50 share	T10 share
2007	0.745 [0.727, 0.763]	0.042 [0.034, 0.051]	0.594 [0.569, 0.618]
2011	0.729 [0.714, 0.745]	0.047 [0.039, 0.054]	0.572 [0.550, 0.594]
2014	0.746 [0.721, 0.771]	0.034 [0.022, 0.047]	0.576 [0.551, 0.602]
2017	0.720 [0.706, 0.734]	0.049 [0.041, 0.057]	0.557 [0.540, 0.573]
2021	0.698 [0.682, 0.715]	0.056 [0.048, 0.065]	0.535 [0.513, 0.557]
2024	0.766 [0.747, 0.785]	0.012 [0.003, 0.022]	0.595 [0.570, 0.619]

Table S1: Six-wave EFH income and signed-wealth estimates

Entries are point estimates with 95 percent intervals in brackets. Income excludes imputed owner rent; net wealth is assets less debts and retains negative values. Intervals combine replicate or qualified bootstrap variance with Rubin multiple-imputation variance.

The repository records the analytical status of each result. It labels the 2017–2021 income fall, the 2021–2024 income reversal, and the 2021–2024 wealth increase as primary results already known when the paper was recast. It labels the transfer-composition and survivor-income exercises as descriptive revision hypotheses; the survivor-wealth, wealth-component, income-convention, upper-tail, wealth-definition, and joint-rank exercises as post-analysis descriptive checks; and the endpoint and lineage statements as descriptive results. This is a record against post-rewrite claim drift, not a claim that the study was prospectively registered before the data were observed.

<i>Income</i>				
Comparison	Change	Independent 95 pct. CI	Unknown-covariance 95 pct. CI	Holm robust
2007 to 2011	-0.021	[-0.042, -0.001]	[-0.050, 0.007]	No
2011 to 2014	0.002	[-0.018, 0.022]	[-0.025, 0.030]	No
2014 to 2017	-0.015	[-0.036, 0.005]	[-0.044, 0.013]	No
2017 to 2021	-0.079	[-0.096, -0.062]	[-0.103, -0.055]	Yes
2021 to 2024	0.084	[0.052, 0.115]	[0.043, 0.125]	Yes

<i>Net wealth</i>				
Comparison	Change	Independent 95 pct. CI	Unknown-covariance 95 pct. CI	Holm robust
2007 to 2011	-0.016	[-0.040, 0.008]	[-0.050, 0.018]	No
2011 to 2014	0.016	[-0.013, 0.046]	[-0.024, 0.057]	No
2014 to 2017	-0.026	[-0.054, 0.003]	[-0.065, 0.014]	No
2017 to 2021	-0.022	[-0.044, 0.000]	[-0.053, 0.009]	No
2021 to 2024	0.068	[0.043, 0.093]	[0.033, 0.103]	Yes

Table S2: Adjacent-wave EFH Gini contrasts

Changes are later minus earlier adjacent-wave Ginis. The conservative interval allows any feasible cross-wave covariance and uses a conservative degrees-of-freedom bound. The final column also applies Holm adjustment across all ten income and wealth contrasts.

Convention	EFH	CASEN	Difference in changes
Households, excluding rent	0.004	-0.018	0.022 [-0.011, 0.056]
Represented persons, excluding rent	0.001	-0.019	0.020 [-0.012, 0.053]
Households, including rent	-0.003	-0.022	0.018 [-0.014, 0.050]

Table S3: EFH and CASEN income trends under alternative conventions

Entries are 2024 minus 2017 Gini changes. The final column reports the EFH change minus the CASEN change with an independent-survey 95 percent interval. Represented-person and including-rent results are post-analysis descriptive checks and do not enter the primary hypothesis family.

Resource	Cap	Period	Gini change and conservative 95 pct. interval
Effective income	p99.0	2017–2021	-0.074 [-0.098, -0.049]
Effective income	p99.5	2017–2021	-0.077 [-0.101, -0.052]
Effective income	p99.0	2021–2024	0.067 [0.041, 0.093]
Effective income	p99.5	2021–2024	0.069 [0.042, 0.097]
Signed non-pension wealth	p99.0	2021–2024	0.075 [0.037, 0.112]
Signed non-pension wealth	p99.5	2021–2024	0.072 [0.035, 0.109]

Table S4: Upper-tail influence sensitivity

Each outcome is capped at its own weighted percentile in each wave, implicate, and replicate. These influence checks do not replace the uncapped primary estimands.

Definition	Cap	2021 Gini	2024 Gini	Change and conservative 95 pct. interval
Wave-specific provider total	None	0.698	0.766	0.068 [0.033, 0.103]
Wave-specific provider total	p99.0	0.680	0.755	0.075 [0.037, 0.112]
Wave-specific provider total	p99.5	0.689	0.761	0.072 [0.035, 0.109]
Modern-common (2024 crypto excluded)	None	0.698	0.766	0.068 [0.033, 0.103]
Modern-common (2024 crypto excluded)	p99.0	0.680	0.755	0.075 [0.037, 0.112]
Modern-common (2024 crypto excluded)	p99.5	0.689	0.761	0.072 [0.034, 0.109]
Legacy-common (account savings and 2024 crypto excluded)	None	0.701	0.768	0.067 [0.031, 0.102]
Legacy-common (account savings and 2024 crypto excluded)	p99.0	0.683	0.757	0.074 [0.036, 0.111]
Legacy-common (account savings and 2024 crypto excluded)	p99.5	0.692	0.762	0.070 [0.033, 0.108]

Table S5: Wealth-definition sensitivity, 2021–2024

Modern-common removes cryptoassets newly included in the 2024 provider total. Legacy-common additionally removes separately measured account savings in both waves. Caps are recomputed in every implicate and replicate. Intervals use the covariance-agnostic bound.

<i>A. Levels</i>				
Component	Mean share 2021	Mean share 2024	Contribution 2021	Contribution 2024
Principal residence	0.748	0.686	0.457	0.449
Other real estate	0.270	0.317	0.221	0.266
Vehicles and other real assets	0.080	0.076	0.037	0.036
Financial and transaction-account assets	0.076	0.089	0.055	0.073
Debt subtraction	-0.173	-0.169	-0.072	-0.057

<i>B. 2021–2024 change decomposition</i>			
Component	Contribution change	Share term	Concentration term
Principal residence	-0.007 [-0.063, 0.049]	-0.039	0.032
Other real estate	0.044 [-0.017, 0.106]	0.039	0.005
Vehicles and other real assets	-0.001 [-0.014, 0.012]	-0.002	0.000
Financial and transaction-account assets	0.017 [-0.011, 0.046]	0.010	0.007
Debt subtraction	0.015 [-0.013, 0.043]	0.002	0.013

Table S6: Signed non-pension wealth-component accounting, 2021–2024

Components sum exactly to signed non-pension net wealth; mean shares therefore sum to one and Gini contributions sum to the signed wealth Gini. Debt is entered as a negative component. Brackets give covariance-agnostic 95 percent bounds for each 2021–2024 contribution change. The share and concentration terms form an exact symmetric decomposition of each contribution change. All estimates use all 30 implicates and the 1,000 provider replicates in each wave. Company participations remain within financial assets and business machinery within other real assets because EFH does not expose a compatible separate business-wealth aggregate in both waves. The account is descriptive, not causal.

Resource	Domain and weights	2021 Gini	2024 Gini	Change
Effective income	Repeated cross-section	0.439 [0.427, 0.451]	0.523 [0.494, 0.552]	0.084 [0.043, 0.125]
Effective income	Survivors, common 2024 weights	0.444 [0.425, 0.462]	0.528 [0.469, 0.587]	0.084 [0.007, 0.162]
Effective income	Survivors, common 2021 weights	0.440 [0.422, 0.458]	0.536 [0.480, 0.591]	0.096 [0.022, 0.170]
Signed non-pension wealth	Repeated cross-section	0.698 [0.682, 0.715]	0.766 [0.747, 0.785]	0.068 [0.033, 0.103]
Signed non-pension wealth	Survivors, common 2024 weights	0.711 [0.684, 0.738]	0.747 [0.722, 0.771]	0.036 [-0.015, 0.088]
Signed non-pension wealth	Survivors, common 2021 weights	0.709 [0.680, 0.738]	0.741 [0.716, 0.766]	0.032 [-0.022, 0.086]

Table S7: Income and wealth survivor-domain robustness

Each survivor row uses the stated year's final weight and all 1,000 corresponding provider replicates for both years. Change intervals use the covariance-agnostic Cauchy bound because cross-wave imputations were not jointly drawn. Wealth tail shares also move toward greater concentration under both common-weight systems, but their conservative change intervals include zero. This is a 2024 survivor domain, not a population panel.

2 Historical CASEN-lineage benchmark

Main-paper Figure 2 plots the non-interpolated World Bank PIP observations used to summarize the established long-run account. The frozen series is derived from CASEN through the Latin American harmonization pipeline and distributes household per-capita income among persons. It is therefore evidence about the direction and scale of the CASEN-lineage decline, not an independent survey replication or a level match to EFH. Within its internally comparable 2006–2024 spell, the Gini falls from 0.477 to 0.430.

3 Estimands and sample scope

The primary EFH population is households represented by the six national- urban waves in 2007, 2011, 2014, 2017, 2021, and 2024. The primary income resource is monthly effective household income, excluding imputed owner rent. Primary wealth is signed non-pension household net wealth, assets less debts. All 30 provider implicates are retained. Monetary-level changes over time are outside scope because a harmonized price-basis extension is not part of the registered design.

The common EFH–CASEN vector in 2011, 2017, and 2024 contains:

1. urban households, income excluding imputed rent;
2. represented urban persons, household income split equally among all members and household weights multiplied by household size; and
3. urban households, income including imputed rent.

CASEN-only estimates for national coverage, broad equal-split adults, and autonomous income diagnose unmatched choices. They are not presented as EFH adjustments and cannot decompose survey bias.

CASEN 2020 and 2022 enter the income path and component account as non-overlap context. The direct EFH–CASEN comparison remains limited to 2011, 2017, and 2024.

4 Weighted signed Gini

For observations x_i and positive weights w_i , with positive weighted mean μ , the estimator is

$$\widehat{G}(x, w) = \frac{\sum_i \sum_j w_i w_j |x_i - x_j|}{2(\sum_i w_i)^2 \mu}.$$

Negative observations remain in the ordering and numerator. The estimator is undefined if the weighted mean is nonpositive. The production implementation uses the algebraically equivalent sorted-data formula and is tested against the pairwise definition on artificial signed data.

For the zero-cap sensitivity, x_i is replaced by $\max(x_i, 0)$. This changes both the denominator and the pairwise distances. It also maps every negative balance sheet to the same point, which is why the signed distribution remains the primary concept.

5 Multiple imputation and joint covariance

Let $\hat{\theta}_m$ be the vector of three comparability Ginis in implicate m , and let $\widehat{\mathbf{U}}_m$ be its within-implicate replicate covariance. With $M = 30$,

$$\begin{aligned}\bar{\theta} &= M^{-1} \sum_m \hat{\theta}_m, \\ \bar{\mathbf{U}} &= M^{-1} \sum_m \widehat{\mathbf{U}}_m, \\ \mathbf{B} &= (M - 1)^{-1} \sum_m (\hat{\theta}_m - \bar{\theta})(\hat{\theta}_m - \bar{\theta})', \\ \mathbf{T} &= \bar{\mathbf{U}} + (1 + M^{-1})\mathbf{B}.\end{aligned}$$

Every linear contrast $\mathbf{c}'\bar{\theta}$ therefore uses $\mathbf{c}'\mathbf{T}\mathbf{c}$. This preserves the covariance created by estimating all conventions on the same replicate draws. Scalar Rubin degrees of freedom are computed for each reported contrast from its projected within- and between-imputation variance.

The 2017, 2021, and 2024 EFH waves use all 1,000 provider replicate weights. The 2007, 2011, and 2014 waves lack public provider replicates; their intervals use a fixed-seed bootstrap within the released income strata and remain qualified. The 2011 EFH ladder uses common bootstrap draws across its three components.

6 Income–wealth rank overlap

The joint-distribution exercise compares households' positions in the income and signed non-pension net-wealth distributions within each EFH wave. Let R_i^I and R_i^W denote the weighted income and wealth midranks of household i . The reported Spearman coefficient is the weighted Pearson correlation of these two rank variables.

For a tail containing population share q , let $T_i^I(q)$ and $T_i^W(q)$ denote fractional membership in the upper income and wealth tails, and let $B_i^I(q)$ and $B_i^W(q)$ denote the corresponding lower-tail memberships. The conditional overlap estimators are

$$\hat{p}_T(q) = \frac{\sum_i w_i T_i^I(q) T_i^W(q)}{\sum_i w_i T_i^I(q)}, \quad \hat{p}_B(q) = \frac{\sum_i w_i B_i^I(q) B_i^W(q)}{\sum_i w_i B_i^I(q)}.$$

The main paper reports $\hat{p}_T(0.20)$, $\hat{p}_B(0.20)$, and $\hat{p}_T(0.10)$. Under independent income and wealth ranks, each conditional probability equals its tail's population share: 0.20 for either quintile and 0.10 for the top decile.

Ranks and tail boundaries are recomputed separately in every impute and replicate. When a set of tied observations crosses a boundary, fractional membership allocates exactly the required weighted population share without arbitrary ordering inside the tie. Each statistic is then pooled over the 30 imputates using Rubin's rules. These quantities describe same-wave association; they do not track particular households over time or identify why income and wealth positions overlap. These statistics are descriptive extensions and do not enter the registered adjacent-wave testing family.

7 CASEN complex-design covariance

CASEN uses the released household expansion weight, pseudo-stratum, and ultimate-cluster variables. The 2011 design additionally applies the released first-stage finite-population correction. The consolidated 2017, 2020, 2022, and 2024 files do not release a comparable FPC and are estimated under the with-replacement ultimate-cluster convention.

For each target, the code computes the exact weighted-Gini influence function. For represented persons, household influence is multiplied by household size and divided by its base-weighted mean, so the linearized household design targets the person-weighted Gini. The three influence functions are stacked in one survey mean, producing a source covariance matrix rather than three independent standard errors. Direct tests require every vector point and standard error to equal the corresponding scalar design estimator.

EFH and CASEN are treated as independent surveys. The covariance of their gap vector is therefore the sum of their source covariance matrices. This is a declared assumption, not an empirical estimate of cross-source covariance. The nine convention-by-year gaps and six changes from the primary convention are reported as a descriptive sensitivity family. Holm-adjusted values are retained for transparency, but the family was examined after the primary analysis and carries no confirmatory claim.

Table S8 records the fieldwork facts that matter for temporal comparison. Tables S9 and S10 then report the full concept ladder and the primary same-year estimates.

8 Income-component accounting

Total income is divided into labor, public transfers, and other autonomous income. Every household is ranked by total income, and tied totals receive a common weighted midrank. If s_k is component k 's mean

Source	Year	Urban households	Field period	Mode	Income reference	Response
EFH	2017	4,549	20 Jun.–5 Dec. 2017	In person, electronic	Current monthly; labor last month	n.r.
EFH	2021	4,400	4 Sep. 2021–31 Mar. 2022	In-person precontact; mainly telephone interview	Current monthly; labor last month	n.r.
EFH	2024	4,649	22 Jul. 2024–13 Jan. 2025	In person, electronic	Current monthly; labor last month	71.5%
CASEN	2011	46,561	17 Oct. 2011–22 Jan. 2012	In person	Monthly monetary; labor last month	80.0%
CASEN	2017	57,460	2 Nov. 2017–4 Feb. 2018	In person, CAPI	Monthly monetary; labor last month	75.5%
CASEN	2020	49,777	Nov. 2020–Jan. 2021	Sequential mixed; mainly telephone	Monthly monetary; labor last month	64.0%
CASEN	2022	57,130	Nov. 2022–Jan. 2023	In person, CAPI	Monthly monetary; labor last month	70.3%
CASEN	2024	64,432	1 Nov. 2024–2 Feb. 2025	In person, CAPI	Monthly monetary; labor last month	75.8%

Table S8: Fieldwork and analytical-sample crosswalk. Household counts refer to the analysis files and, for CASEN, the urban household-head domain. Broad mode labels and response rates reproduce released survey documentation and are not used as explanatory variables. “n.r.” means that the reviewed wave document did not report an overall response rate; target attainment is not substituted. Exact dates are shown when the reviewed documentation provides them, and month ranges otherwise.

<i>Households, excluding rent</i>						
Year	EFH	CASEN	EFH minus CASEN	Change from primary gap	Holm p	
2011	0.532	0.489	0.043 [0.027, 0.059]	Reference	0.000	
2017	0.519	0.482	0.037 [0.021, 0.052]	Reference	0.000	
2024	0.523	0.464	0.059 [0.030, 0.089]	Reference	0.000	
<i>Represented persons, excluding rent</i>						
Year	EFH	CASEN	EFH minus CASEN	Change from primary gap	Holm p	
2011	0.521	0.499	0.022 [0.005, 0.039]	-0.021 [-0.031, -0.012]	0.012	
2017	0.528	0.488	0.040 [0.024, 0.055]	0.003 [-0.005, 0.011]	0.000	
2024	0.529	0.470	0.060 [0.031, 0.088]	0.001 [-0.009, 0.010]	0.000	
<i>Households, including rent</i>						
Year	EFH	CASEN	EFH minus CASEN	Change from primary gap	Holm p	
2011	0.497	0.453	0.044 [0.028, 0.060]	0.001 [-0.003, 0.005]	0.000	
2017	0.478	0.439	0.039 [0.024, 0.054]	0.002 [-0.001, 0.005]	0.000	
2024	0.475	0.418	0.057 [0.029, 0.085]	-0.002 [-0.007, 0.002]	0.000	

Table S9: EFH–CASEN comparability ladder

Entries are jointly estimated Ginis and covariance-aware 95 percent intervals. Within-source covariance is retained; EFH and CASEN are assumed independent. Gap changes are estimand sensitivities, not a decomposition of survey bias.

Year	EFH income Gini	CASEN income Gini	EFH minus CASEN
2011	0.532 [0.521, 0.543]	0.489 [0.477, 0.500]	0.043 [0.027, 0.059]
2017	0.519 [0.507, 0.531]	0.482 [0.472, 0.491]	0.037 [0.021, 0.052]
2024	0.523 [0.494, 0.552]	0.464 [0.458, 0.469]	0.059 [0.030, 0.089]

Table S10: Closest-concept same-year EFH and CASEN estimates

Both series use urban households and exclude imputed owner rent, but questionnaire, income construction, survey mode, and sample design still differ. CASEN intervals use released strata, PSU, weights, and the 2011 finite-population correction; difference intervals treat the two surveys as independent. The gap is observed, not a decomposition of source bias.

Measure	Comparison	EFH	CASEN	EFH minus CASEN
Bottom-50 income share	2011	0.161 [0.155, 0.167]	0.183 [0.177, 0.189]	-0.022 [-0.031, -0.014]
Bottom-50 income share	2017	0.164 [0.157, 0.170]	0.187 [0.183, 0.192]	-0.023 [-0.031, -0.016]
Bottom-50 income share	2024	0.163 [0.152, 0.175]	0.195 [0.192, 0.197]	-0.032 [-0.043, -0.020]
Bottom-50 income share	2017–2024 change	-0.001 [-0.014, 0.013]	0.008 [0.002, 0.013]	-0.008 [-0.022, 0.006]
Top-10 income share	2011	0.416 [0.404, 0.427]	0.376 [0.363, 0.390]	0.039 [0.022, 0.057]
Top-10 income share	2017	0.397 [0.383, 0.411]	0.371 [0.360, 0.382]	0.026 [0.008, 0.043]
Top-10 income share	2024	0.399 [0.364, 0.435]	0.350 [0.343, 0.356]	0.050 [0.013, 0.086]
Top-10 income share	2017–2024 change	0.003 [-0.036, 0.041]	-0.021 [-0.034, -0.009]	0.024 [-0.016, 0.064]

Table S11: Income shares at the bottom and top of the EFH and CASEN distributions

Shares are fractions of total household income; intervals are 95 percent intervals. Level gaps and change rows use independent-survey or independent-wave standard errors. The full covariance-agnostic change bounds are retained in the public CSV. A rising bottom-50 share and falling top-10 share indicate compression at the respective tails.

share and C_k its concentration coefficient, the implementation verifies

$$G = \sum_k s_k C_k.$$

The exact symmetric change identity is

$$\Delta(s_k C_k) = \frac{1}{2} \Delta s_k (C_{kt} + C_{k0}) + \frac{1}{2} \Delta C_k (s_{kt} + s_{k0}).$$

Both identities are checked symbolically in SymPy and numerically against an independent pairwise Gini over at least 120 artificial distributions.

EFH estimates use each of the 30 implicates, provider replicates from 2017 onward, and Rubin pooling. CASEN component uncertainty uses deterministic stratified delete-one-PSU JK_n replicates. Replicates are constructed from the full released design before evaluating the urban domain, which preserves strata that contain both urban and nonurban PSUs. The 2011 JK_n applies the released stratum sampling fraction. The corrected transfer component is defined residually so the three components add exactly to corrected monetary income. Its difference from the direct subsidy field is recorded rather than silently rounded.

Table S12 gives the complete component account used by the main-paper figure and endpoint discussion.

CASEN					
Year	Component	Mean share	Concentration	Gini contribution	Change contribution
2017	Labor income	0.847	0.525	0.445	Reference
2017	Other autonomous income	0.128	0.321	0.041	Reference
2017	Public transfers	0.025	-0.165	-0.004	Reference
2020	Labor income	0.775	0.597	0.463	0.018
2020	Other autonomous income	0.175	0.351	0.062	0.020
2020	Public transfers	0.049	-0.138	-0.007	-0.003
2022	Labor income	0.824	0.527	0.434	-0.029
2022	Other autonomous income	0.128	0.346	0.044	-0.017
2022	Public transfers	0.047	-0.132	-0.006	0.001
2024	Labor income	0.822	0.515	0.423	-0.011
2024	Other autonomous income	0.128	0.377	0.048	0.004
2024	Public transfers	0.050	-0.154	-0.008	-0.001
EFH					
Year	Component	Mean share	Concentration	Gini contribution	Change contribution
2017	Labor income	0.822	0.538	0.442	Reference
2017	Other autonomous income	0.170	0.465	0.079	Reference
2017	Public transfers	0.008	-0.254	-0.002	Reference
2021	Labor income	0.749	0.524	0.392	-0.050
2021	Other autonomous income	0.116	0.349	0.040	-0.039
2021	Public transfers	0.135	0.051	0.007	0.009
2024	Labor income	0.843	0.561	0.473	0.081
2024	Other autonomous income	0.126	0.439	0.056	0.015
2024	Public transfers	0.031	-0.181	-0.006	-0.012

Table S12: Income-component shares, concentrations, and Gini contributions

Households are ranked by total income. Each year's component contributions add to its total-income Gini. Changes use the exact symmetric share and concentration decomposition and are descriptive, not causal. CASEN 2020 and 2022 provide non-overlap context; primary same-year EFH comparisons remain 2011, 2017, and 2024.

9 Wealth-component accounting

The 2021 and 2024 files permit an unchanged five-part identity for signed non-pension net wealth: principal residence, other real estate, vehicles and other real assets, financial and transaction-account assets, and debt subtraction. Debt enters with a negative sign. With households ranked by total signed net wealth, the same concentration-index and symmetric-change identities used for income make the component contributions add exactly to each wave’s wealth Gini and to its 2021–2024 change.

All 30 implicates and 1,000 provider replicates are retained in each wave. Change intervals use the covariance-agnostic bound because the cross-wave imputations were not jointly drawn. The implementation verifies the household wealth identity to 0 CLP and the Gini add-up identity to numerical precision, and independently reproduces the identities in 120 artificial signed balance-sheet distributions that include negative net wealth. Company participations remain within financial assets and business machinery within other real assets; the released aggregates do not permit a stable separate business-wealth component. Table S6 reports the resulting descriptive account.

10 EFH survivor-domain protocol

The released 2024 predecessor key identifies 1,801 unique households from the 2021 refresh cohort. Linkage must be one-to-one and complete in the two imputed household files and both replicate-weight files. Both years first use the 2024 final weight and provider replicates, then use common 2021 weights and replicates as a second specification. The 30 implicates are estimated separately by year for the income and signed non-pension wealth Ginis and tail shares. No cross-wave implicate pairing is assumed. Change intervals use the Cauchy upper bound on the standard error because joint cross-wave imputation draws are unavailable.

The attrition audit compares survivors with the 1,054-household non-reinterviewed remainder of the 2021 refresh cohort on weighted baseline income, signed wealth, household size, macrozone, and stratum. Standardized differences are reported for every characteristic. Private support counts must exceed the disclosure threshold in every categorical row. The output contains aggregates only.

Table S13 reports the continuous baseline balance statistics; the aggregate validation record also retains the categorical macrozone and stratum checks.

Baseline	Survivors: estimate (SE)	Not reinterviewed: estimate (SE)	Std. difference
Effective income (million CLP)	1.565 (0.041)	1.568 (0.060)	-0.00
Signed non-pension wealth (million CLP)	80.69 (4.59)	74.22 (4.60)	0.04
Household size	3.17 (0.05)	3.04 (0.06)	0.09

Table S13: Baseline balance in the 2021 refresh cohort

Survey-weighted 2021 means compare the 1,801 households observed again in 2024 with the 1,054-household non-reinterviewed remainder of the 2021 refresh cohort. Monetary entries are millions of nominal 2021 CLP; standard errors are in parentheses. This comparison describes observed baseline balance; it cannot rule out selection on unobserved characteristics or subsequent outcomes.

11 Synthetic validation contract

The primary EFH family, closest-concept CASEN comparison, component account, and survivor exercise pass their named validation gates. A prospective screen for a wider measurement-extension family failed and remains failed, so the signed lower-tail family is excluded. Imputed-rent and represented-person estimates are retained only as post-analysis descriptive checks. The full decision history, thresholds, seeds, and immutable failed artifacts are recorded in [analysis/validation/MEASUREMENT_EXTENSION_FALLBACK.md](#); keeping that history in the repository prevents it from overwhelming the reader-facing supplement.

The separate canonical EFH Monte Carlo uses seed 20260817 and 1,500 repetitions per scenario, or 250 per wave-scenario design cell. It validates the income mean, signed income Gini, p99-capped income Gini, and signed net-wealth Gini under complete data and correctly specified missing-at-random imputation. All fixed gates pass; the adverse missing-not-at-random income control fails visibly as required.

The income-component and survivor-domain analyses are separate modules. Their point identities are verified symbolically and against pairwise Gini and concentration formulas in 120 artificial distributions.

EFH components use provider replicates and Rubin pooling; CASEN components use deterministic JK_n replicates, with their summed Gini and standard error checked against the independent Taylor implementation in every wave. The five-wave CASEN core-design Monte Carlo uses 250 repetitions per DGP and the frozen bias, SE-ratio, and coverage thresholds. Across 31,250 estimator-level evaluations, the maximum absolute identified-coefficient bias is 0.00049, SE ratios range from 0.885 to 1.086, and coverage ranges from 0.904 to 0.976; all gates pass. A separate 250-repetition-per-wave audit requires an explicitly unweighted estimator to fail visibly. These checks do not convert descriptive component accounting into a causal design.

12 Extended design and quality diagnostics

Tables S14 and S15 summarize the data and validation gates. Tables S16 and S17 collect the detailed distributional and component endpoint checks.

Wave	N	Weighted N	M	Zero income (pct.)	Negative wealth (pct.)	Variance	Accounting
2007	3,828	3,847,952	30	2.27	13.79	Qualified approximation	Pass
2011	4,059	4,233,502	30	2.41	16.67	Qualified approximation	Pass
2014	4,502	4,701,109	30	1.13	15.08	Qualified approximation	Pass
2017	4,549	4,868,518	30	1.05	12.76	Provider 1,000 replicates	Pass
2021	4,400	5,616,257	30	0.98	10.44	Provider 1,000 replicates	Pass
2024	4,649	6,693,423	30	2.79	11.31	Provider 1,000 replicates	Pass

Table S14: Wave-level sample and validation diagnostics

Target population is national-urban households. Weighted counts use the household expansion factor. Completed-data weighted missingness is exactly zero for primary effective income and net wealth in every wave. All waves contain 30 implicates. Early-wave interval estimates are qualified because provider replicate weights are unavailable.

12.1 Descriptive 2007–2024 endpoint

Across the full period, the 2024 EFH household effective-income Gini is 0.030 below the 2007 point estimate. Its conservative 95% interval, $[-0.076, 0.016]$, ranges from a larger decline to a small increase. The endpoint comparison therefore suggests modest compression but does not precisely establish its size.

The endpoint is generated separately from the ten registered adjacent-wave comparisons. Its point estimate is the 2024 level minus the 2007 level. The independent-wave calculation is retained for transparency, but the manuscript uses the Cauchy–Schwarz worst-case standard error because the cross-wave covariance is not released. No equivalence margin was prospectively registered, so an interval containing zero is not evidence that the endpoints are equivalent.

13 External-source lineage

The versioned lineage ledger distinguishes publishers from upstream survey systems. Chilean SEDLAC, ECLAC/BADEHOG, PovcalNet/PIP, LIS income, and OECD IDD observations derive from CASEN, although the transformations, statistical units, and resource definitions differ. OECD WDD and LIS/LWS wealth observations derive from EFH. These relationships show robustness within a survey lineage, not independent replication across Chilean household surveys.

WIID is retained as a secondary aggregator whose records can inherit several of those sources. WID instead combines survey, fiscal, administrative, and macroeconomic information under versioned distributional-national-accounts methods. UBS/Credit Suisse estimates modelled country-level adult wealth from surveys, balance sheets, regressions, and distributional assumptions whose inputs can vary by edition. Those constructions provide concept-labelled context rather than additional observations of the EFH household estimand.

Every arrow in the manuscript lineage diagram is backed by an official source note, retrieval date, and SHA-256 in `analysis/config/chile_income_source_lineage.csv`. Shared upstream microdata do not imply identical methodology or identical published values.

Table S18 records the permitted use of every audited source family.

Check	Status	Consequence
Estimator unit tests	Pass	Core weighted estimators satisfy synthetic identities and regression tests.
Symbolic and independent numerical identities	Pass	SymPy algebra and 120 independent weighted-Gini comparisons pass.
Synthetic Monte Carlo	Pass	At seed 20260817, 1,500 repetitions per scenario (250 per wave-scenario design cell) meet income-mean, signed-income-Gini, p99-capped-income-Gini, and signed-net-wealth-Gini bias, SE-ratio, and coverage gates; adverse MNAR failure is visible for income.
WID archive schema and partitions	Pass	Only normalized Ginis and shares are available for source-specific context.
Income accounting identity	Pass	Coverage and residual tolerances pass in every wave.
2024 provider benchmark	Pass	The published 2024 mean, median, and combined standard error are reproduced at the registered tolerances.
2017 published-SE convention	Qualified	Guide convention gives 22,671.092 (published 22,671); the harmonized R-denominator/base-point Rubin SE is 22,715.546, 0.1965 percent higher.
2017/2021/2024 variance	Pass	All 1,000 provider replicate weights enter Rubin-combined uncertainty.
2007/2011/2014 variance	Qualified	Intervals use a household-within-income-stratum approximation, not a fully design-valid variance.
Monetary-level trends	Excluded	No CPI, FX, PPP, or nominal-level connection is reported.

Table S15: Validation and robustness summary

Pass means the named automated check ran successfully in this build. Qualified and excluded are not passes.

Resource	Statistic	2021–2024 change and conservative 95 pct. interval
Income	Bottom-50 share	-0.049 [-0.068, -0.030]
Income	Top-10 share	0.068 [0.020, 0.115]
Wealth	Bottom-50 share	-0.044 [-0.062, -0.026]
Wealth	Top-10 share	0.060 [0.014, 0.106]
Wealth	Zero-capped wealth Gini	0.056 [0.025, 0.088]

Table S16: Distributional-share and zero-cap robustness

All changes use the same covariance-agnostic bound as the headline Ginis. The share movements and the zero-capped wealth Gini corroborate the direction of the 2021–2024 results.

Source	Component	Contribution change	Share term	Concentration term
EFH	Labor income	0.031 [-0.006, 0.068]	0.012	0.020
EFH	Other autonomous income	-0.023 [-0.048, 0.001]	-0.020	-0.004
EFH	Public transfers	-0.003 [-0.006, -0.001]	-0.005	0.001
CASEN	Labor income	-0.021 [-0.035, -0.008]	-0.013	-0.009
CASEN	Other autonomous income	0.007 [0.001, 0.013]	-0.000	0.007
CASEN	Public transfers	-0.004 [-0.004, -0.003]	-0.004	0.000

Table S17: Income-component endpoint changes and uncertainty

Each row decomposes the 2017–2024 change in a component’s Gini contribution. Brackets give covariance-agnostic 95 percent bounds formed from the two endpoint standard errors. The three component changes add to the source’s total Gini change; the two margins add exactly to each component change.

Source	Domain	Unit	Decision	Principal obstacle or lineage
WID	income and wealth	Equal-split adult	Concept-labelled context	Population unit income boundary wealth boundary national-account alignment top-tail adjustment and vintage all differ from EFH
UBS/Credit Suisse	wealth	Adult	Concept-labelled context	Modeled adult concept differs from EFH household survey concept; edition revisions can dominate apparent time changes; no comparable income series
CASEN	income	Household or person depending on statistic	Harmonized income comparison	Different survey purpose questionnaire income construction population coverage and sampling; CASEN does not measure a comparable balance sheet
SEDLAC	income	Household per-capita person	Concept-labelled context	Shared CASEN microdata do not imply identical processing; per-capita unit imputed rent and national coverage differ from EFH
LIS income	income	Equivalized person or household as labelled	Concept-labelled context	CASEN lineage is not independent survey replication; equivalence scale and disposable-income boundary differ from EFH
OECD income	income	Equivalized person	Concept-labelled context	Shared CASEN source; taxes transfers equivalence scale population and reference year differ from EFH
Povcal-Net/PIP	income	Household per-capita person	Concept-labelled context	CASEN lineage is not independent corroboration; national scope per-capita welfare and harmonization differ from EFH
ECLAC	income	Person or household depending on table	Concept-labelled context	CASEN lineage is not independent corroboration; regional transformations need not match the EFH household estimand
WIID	income	Source-specific	Concept-labelled context	An aggregator is not a single harmonized survey and duplicate ancestry may appear as multiple records
SWIID	income	Standardized person-level concept	Concept-labelled context	Modeled standardization is not independent household-survey replication and may inherit shared microdata
INE ESI	labor income	Person for current published income indicators	Concept-labelled context	Beginning in 2021 the survey stopped collecting other household income sources and publishing household-level income estimates
U. Chile EOD	income	Household or person as defined by study	Concept-labelled context	Geographic scope and income construction differ from the national EFH-CASEN comparison
OECD wealth	wealth	Household	Concept-labelled context	EFH-derived values are not independent survey corroboration; harmonized categories and vintages may differ
LIS/LWS wealth	wealth	Household	Concept-labelled context	EFH lineage is not independent wealth-survey replication; category construction may differ
National accounts	aggregate income and wealth	Institutional sector or national aggregate	Not distributionally comparable	No household-level distribution; institutional-sector boundary and valuation differ from EFH
INE EPF	household income and expenditure	Household	Concept-labelled context	Study-specific applications are not recurring international income-Gini series; population coverage timing questionnaire and income construction differ; no 2024 observation
EPS	wealth	Person-weighted mixed respondent and household scope	Lineage audit only	Person rather than household weighting; mixed ownership unit; high and changing wealth-value nonresponse; incomplete mortgage and business values; 2015 design fields unavailable; screening trend is missing-value-sensitive

Table S18: External-source comparability decisions

CASEN is the only public microdata source admitted to a directly estimated EFH income comparison. SEDLAC, ECLAC, PovcalNet/PIP, LIS income, and OECD IDD are CASEN-lineage transformations or disseminations; OECD WDD and LIS/LWS wealth are EFH-lineage. EPS independently measures wealth but mixes person and household scopes; its quantitative trend was withheld after the fail-closed microdata screen. ESI, EPF, and the University of Chile EOD use independent microdata but do not supply the same national-urban household-total-income path through 2024. WIID, SWIID, WID, UBS/Credit Suisse, and national accounts serve distinct aggregation, standardization, integration, modeling, or macro-account roles.

14 Reproduction and disclosure record

Authorized EFH users can run the complete analysis from the project root after placing provider files in the documented layout or pointing the build wrapper to another secure directory. The public-input path rebuilds CASEN and external context only. CI compiles both documents from the reviewed public aggregate bundle and does not access EFH microdata.

The public release bundle excludes rows, identifiers, exact restricted-input hashes, private diagnostics, and support counts. Banco Central de Chile does not endorse the paper, code, or conclusions. Provider correspondence remains separate from the automated statistical gates.

15 Limitations inventory

The analysis is descriptive and repeated cross-sectional. It cannot identify the causes of cross-wave changes; survey bias; causal effects of transfers, housing policy, pensions, social conflict, or the pandemic; panel mobility; top-tail corrections; or the value of an unobserved population distribution. The early-wave EFH intervals cannot recreate confidential clustering, calibration, or nonresponse adjustments. Cross-source agreement is not validation when one public series inherits the other source's microdata.

Full source references are listed in the main paper; this supplement adds no separate bibliographic claims.